Here is the algorithmic extraction and detailed breakdown for **Batch 6 (Videos 51 through 60)** of the Trade City Pro technical analysis curriculum.
This batch covers the advanced mathematical tuning of oscillators, multi-timeframe scaling factors, divergence and momentum exhaustion systems, and the structural foundation of trade planning.

Video 51: "روش استفاده از اوسیلاتورها پارت 6 | آچارکشی اندیکاتورها و اسیلاتورها 8" (Part 51)

- **Core Concept**: Dynamic parameter recalibration ("Achar-Kesh-e Oscillator") to eliminate market noise and false breakout signals 1, 2.
- **System Problem**: Standard fixed parameters (e.g., RSI 14) generate excessive whipsaws and premature overbought/oversold readings during volatile or choppy consolidation phases 1, 3.
- **Algorithmic Recalibration Logic**:
- **Cycle Period Calculation**:
- Measure the distance in candle count from swing low to the next distinct swing low (or swing high to swing high) within the current market regime 1, 2:
- \\L_{\text{Cycle}} = t_{\text{SwingLow}(n)} - t_{\text{SwingLow}(n-1)}\\
- Compute the rolling average cycle length across recent structural swings 1, 2:
- \\\bar{L}*{\text{Cycle}} = *frac{1}{k}*\sum{i=1}^{k} L_{\text{Cycle}, i}\\
- **Parameter Injection**:
- Replace the arbitrary 14 lookback with \\(\bar{L}_{\text{Cycle}}\\) (e.g., dynamically adjusting RSI from 14 to 22 or 27) 2-4.
- **Regime Shift Governor**:
- Parameters remain valid only within the current market structure (range box or trending channel) 1.
- When price breaks out of the macro range into an aggressive trend, recalculate the baseline cycle length from the new structural swing points 1.

Video 52: "روش استفاده از اوسیلاتورها پارت 7 | راز اندیکاتورها و اوسیلاتور ها 8" (Part 52)

- **Core Concept**: Multi-Timeframe Multiplier Factor (MF) for mathematical sub-window scaling 5, 6.
- **System Problem**: Trading on a 15-minute execution chart while manually checking 1-Hour or 4-Hour oscillators creates execution delays and cognitive overload 5, 7.
- **The Multiplier Factor (MF) Mathematical Engine**:
- To display a higher-timeframe indicator directly on a lower-timeframe execution chart without multi-timeframe indicator redraw bugs 6:
- \\MF = \frac{\text{Higher Timeframe in Minutes}}{\text{Lower Timeframe in Minutes}}\\
- *Example (4H onto 15m)*: \\(\frac{4 \times 60}{15} = \frac{240}{15} = 16\\) 6.
- *Example (Daily onto 15m)*: \\(\frac{24 \times 60}{15} = \frac{1440}{15} = 96\\) 8.
- **Scaling Formula**:
- \\\text{Period}*{\text{Adjusted}} = *text{Period}*{\text{Base}} \times MF\\

- **RSI (Base 14)**: Set to \\(14 \times 16 = 224\\) on the 15m chart to mirror the 4H RSI line 6, 8.
- **MACD (12, 26, 9)**: Fast = \\(12 \times 16 = 192\\); Slow = \\(26 \times 16 = 416\\); Signal = \\(9 \times 16 = 144\\) 9.
- **Moving Average (e.g., 25 SMA)**: \\(25 \times 16 = 400\\) SMA on the 15m chart tracks the 4H 25 SMA 6.
- **Execution Advantage**: Provides higher-timeframe confirmation (structural trend, inflection breaches, and momentum shifts) on the execution chart without switching tabs 5, 9.

Video 53: "روش استفاده از اوسیلاتورها پارت 8 | آخرین مطلب اندیکاتور و اوسیلاتور 8" (Part 53)

- **Core Concept**: Method 8 — Systematic Divergence classification, Hidden Divergence trend-continuation setups, and master rules for oscillators 10, 11.
- **Algorithmic Rules for Divergence**:
- **Regular Bearish Divergence (RD-)**:
- Price forms Higher High (\\(P_H > P_{\text{prev } H}\\)) while Oscillator forms Lower High (\\(Osc_H < Osc_{\text{prev } H}\\)) 12. Signals upside momentum exhaustion 12.
- **Hidden Bullish Divergence (HD+)**:
- Price forms Higher Low (\\(P_L > P_{\text{prev } L}\\)) while Oscillator prints a deeper Lower Low (\\(Osc_L < Osc_{\text{prev } L}\\)). Signals continuation in the primary uptrend 11.
- **Extreme Zone Validation Constraint**:
- Divergences occurring in neutral territory (e.g., RSI 40–60) are discarded 13.
- A valid divergence requires at least one pivot to register inside Overbought (\\(>70\\)) or Oversold (\\(<30\\)) territory 13.
- **Master System Governance**:
- Oscillators serve strictly as **Alerts and Filters**—never standalone execution triggers 10, 14.
- Trade execution requires a formal price action trigger (horizontal boundary break, Candle Setup, or Stop-Trigger order execution) 10, 15.

Video 54: "می شناسی بنداز دور MACD هرچی از اندیکاتور" (Part 54)

- **Core Concept**: Demystifying MACD as a Moving Average difference engine and trading its momentum cycles 16, 17.
- **Mathematical Anatomy**:
- \\(\text{MACD Line} = \text{EMA}*{12} - *text{EMA}*{26}\\) 17.
- \\(\text{Signal Line} = \text{EMA}_9(\text{MACD Line})\\) 17.
- \\(\text{Histogram} = \text{MACD Line} - \text{Signal Line}\\) 18.
- **Microstructure Insights & Algorithmic Rules**:
- **Zero-Line Cross (Trend Shift)**:
- \\(\text{MACD Line} = 0\\) occurs when \\(\text{EMA}*{12} = *text{EMA}*{26}\\) 17, 18. Crossing above zero confirms the short-term cycle is outpacing the long-term cycle 18.

- **Midline Chop Filter**:
- Signal crossovers occurring near the zero line reflect tightening Moving Averages during sideways chop rather than fresh directional trends; filter these out 19.
- **Histogram Deceleration (Early Momentum Warning)**:
- When the Histogram bars begin contracting (fading color) while price pushes to a new extreme, moving average separation is decelerating 20, 21. Treat this as a signal for profit-taking rather than an immediate reversal trade 20, 22.

Video 55: "روش خودم MOM و ROC اوسیلاتور و اندیکاتور" (Part 55)

- **Core Concept**: Measuring velocity decay using Rate of Change (ROC) and Momentum (MOM) to prevent entering at the exhaustion point of impulse legs 23, 24.
- **System Formulas**:
- \\(\text{MOM} = \text{Price}*t* - *text{Price}*{t-n}\\) 23.
- \\(\text{ROC} = \left(\frac{\text{Price}*t* - *text{Price}*{t-n}}{\text{Price}_{t-n}}\right) \times 100\\) 23, 25.
- **Algorithmic Rules & Systematic Execution**:
- **Exhaustion Ceiling (The Stop-Entry Filter)**:
- Map horizontal historical extreme bands on ROC across a 200-candle window on the 4-Hour timeframe 26-28.
- When 4H ROC hits or exceeds its historical positive ceiling:
- **Disable all Long breakout triggers on the 15-minute timeframe** 28, 29.
- Execute profit taking on open positions 27, 29.
- Price may push higher temporarily, but momentum velocity is depleted, creating elevated risk for a sharp pullback 27, 29.
- **Range Box Breakout Indicator**:
- Draw horizontal boundaries around ROC's choppy baseline oscillations 26, 27.
- When ROC breaks decisively outside this consolidation band, it confirms that market volatility has expanded, validating trend-following setups 26, 29.

Video 56: "بنداز دور MACD و RSI اندیکاتور ADX اندیکاتور" (Part 56)

- **Core Concept**: Directional Movement Index (DMI) and Average Directional Index (ADX) for measuring trend strength and order type selection 30, 31.
- **Indicator Configuration**:
- ADX line (trend strength) + \\(+\text{DI}\\) line (bullish pressure) + \\(-\text{DI}\\) line (bearish pressure) 30.
- **Algorithmic Filter & Order Selection Matrix**:
- **Order Type Confirmation (Stop-Market vs. Candle Close)**:
- *Condition*: \\(+\text{DI}\\) and \\(-\text{DI}\\) are widely separated, and ADX is sloping upward 31.
- *Action*: High-velocity trend expansion confirmed; place **Stop-Buy (Market)** or **Stop-Sell (Market)** orders ahead of breakout levels 31.
- *Condition*: \\(+\text{DI}\\) and \\(-\text{DI}\\) are tangled and overlapping 31.

- *Action*: Do **not** place stop-trigger orders (high fakeout probability); require a confirmed candle body close beyond the boundary before entry 31.
- **ADX Exhaustion Signal**:
- ADX measures trend strength, not direction 32.
- When ADX reaches extreme historical levels (>45–50) and curls downward, trend exhaustion is flagged; tighten trailing stops and execute profit-taking 32-34.

Video 57: "اسناپیری خروج بزن پیوت پوینت" (Part 57)

- **Core Concept**: Multi-timeframe Floor Trader Pivot Points for mathematical profit taking and exit mapping 35, 36.
- **Configuration Parameters**:
- **Calculation Base**: Floor Trader Pivots calculated on the **4-Hour and Weekly** periods (lower-timeframe pivots, such as 15m, generate excessive noise and are discarded) 36.
- Key levels: Central Pivot (\\(P\\)), Resistance levels (\\(R_1, R_2, R_3, R_4\\)), Support levels (\\(S_1, S_2, S_3, S_4\\)) 35, 37.
- **Algorithmic Exit & Target Mechanics**:
- **The Structural Target Ceiling (\\(R_3 / R_4\\) and \\(S_3 / S_4\\))**:
- In standard daily volatility, price rarely breaches \\(R_3\\) or \\(S_3\\) 35.
- \\(R_4\\) (for Longs) and \\(S_4\\) (for Shorts) mark extreme volatility limits 35, 37.
- **Action**: If an active momentum trade reaches \\(R_3\\) or \\(S_3\\) within a single 4-Hour candle, initiate aggressive profit-taking; close remaining positions at \\(R_4 / S_4\\) 35, 37, 38.
- **Directional Asymmetry**:
- Do not use the central \\(P\\) line as a standalone entry signal 37.
- Use breaches beyond \\(R_1\\) (for Longs) or below \\(S_1\\) (for Shorts) as confirmation of directional displacement away from the baseline range 37.

Video 58: "بهترین اندیکاتور ارزدیجیتال | تشخیص زمان ورود به معامله TCP" (Part 58)

- **Core Concept**: Quantitative Market Session Profiling (Tokyo/Asia, London/Europe, New York/US) and session-trend dependencies 39, 40.
- **Session Parameter Architecture**:
- **Asia (Tokyo)**: Midnight UTC to 08:00 UTC (accumulation / manipulation range) 39, 40.
- **Europe (London)**: 07:00 UTC to 16:00 UTC (initial directional displacement) 40.
- **US (New York)**: 12:00 UTC to 21:00 UTC (high-volume momentum follow-through) 40.
- **Session Correlation Engine**:
- **Regime 1 (Trend Continuity)**:
- If London establishes a strong directional trend (Trend Up) and New York confirms (Trend Up), Asia typically trades sideways in a consolidation range (Manipulation / Range) 40.
- *Rule*: Expect trend continuation into the following London/US overlap; set breakout alerts for the London open 40.
- **Regime 2 (London Failure / US Reversal)**:

- If London fails to trend and trades within a choppy range, the primary momentum displacement shifts to the US session open (13:30–14:30 UTC), often driven by macro news flow 40, 41.
- **Execution Restriction**:
- Avoid opening new 15-minute breakout positions during the low-liquidity gap between the US close and the Asia open (21:00 to 00:00 UTC) due to elevated slippage and low taker volume 42.

Video 59: "خروج بزنیم BTC چطوری بیشترین سود از ارزدیجیتال بگیریم با پیر" (Part 59)

- **Core Concept**: Systematic pair trading and Altcoin relative-strength scanning using ALT/BTC charts 43, 44.
- **Quantitative Sieve for Altcoin Selection**:
- When Bitcoin trends upward or stabilizes in a range, evaluate two asset charts before trading any Altcoin:
- **ALT/USDT**: The dollar-denominated chart 43, 44.
- **ALT/BTC**: The Bitcoin-denominated relative strength ratio 43, 44.
- **Selection Decision Matrix**:
- **Pass Condition**: ALT/BTC is breaking out of a consolidation pattern (e.g., falling wedge, horizontal resistance) on expanding volume 43, 44.
- *Action*: Long the Altcoin on ALT/USDT; its returns will outpace Bitcoin's move due to relative-strength leverage 43-45.
- **Reject Condition**: ALT/USDT shows a minor technical breakout, but ALT/BTC remains locked in a downtrend or hugging support 45-47.
- *Action*: Do **not** enter the Altcoin; trade Bitcoin instead 45, 46. Any pullback in BTC/USDT will cause a disproportionate dump in the Altcoin 47.

Video 60: "هرچی از ساخت و نوشتن استراتژی ترید میشناسی بنداز دور بر اساس پرایس اکشن" (Part 60)

1. **Core Concept**: The Written Trading Plan ("Plan-e Maktoob-e Moamelati") as a prerequisite for algorithmic execution and backtesting 48.
2. **Core Rules & Structural Framework**:
3. A strategy cannot be traded or automated until every component is documented in writing with unambiguous mathematical and visual criteria 48.
4. **The 5 Components of the System Document**:
5. *Market Regime & Trend Identification*: Rules defining Bullish, Bearish, or Choppy markets (Dow Theory swing points, 99 SMA orientation, Market Cycle phase) 49-51.
6. *Entry Setup & Trigger Specification*: Candlestick geometry requirements, 7 SMA positioning, order types (Stop-Market trigger coordinates), and buffer tolerances 50.
7. *Secondary Filters*: Multi-timeframe indicator alignment (RSI threshold, volume expansion, order book depth validation) 49, 50.
8. *Invalidation & Stop Placement*: Objective invalidation level where the thesis fails; maximum initial risk percentage (\\(0.25\%\\) to \\(1.0\%\\)) 52.

9. *Trade Management & Profit Ladder*: Quantitative partial take-profit levels (\\(2R, 5R, 10R\\)) and trailing stop progression rules 52.

Progress Tracker

- **Completed Batches**:
- Batch 1: Videos 1–10 (Foundations, Dow Theory, S/R, Capital & Risk Management)
- Batch 2: Videos 11–20 (Market Dominance Ratios, BB on RSI, Risk Expectancy, Profit Ladder)
- Batch 3: Videos 21–30 (Trend Health Analysis, Geometry, Multi-Timeframe Velocity, Volume Dynamics)
- Batch 4: Videos 31–40 (VPA Modeling, 3-SMA System, 15m Candle Setup Engine, Level 2 Order Books)
- Batch 5: Videos 41–50 (Order Book Microstructure, Indicators vs. Oscillators, Methods 1–5 of Oscillator Analysis)
- **Batch 6: Videos 51–60 (Recalibration, MF Multi-Timeframe Scaling, MACD/ROC/ADX Mechanics, Session Timing, Written Plans)**
- **Upcoming**:
- **Batch 7 (Videos 61–70)**: Market Structure & Cycles (HWC/MWC/LWC), Futures Revenue Architecture, Momentum Breakouts, Leverage Mathematics, Pre-Trade Checklists, and Entry Mechanics.

Let me know when you are ready to proceed with **Batch 7 (Videos 61 through 70)**.